

Qualification Pack



Solar Domestic Product Assembler

QP Code: SGJ/Q4902

Version: 1.0

NSQF Level: 2.5

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Qualification Pack

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SGJ/Q4902: Solar Domestic Product Assembler

Brief Job Description

Solar Domestic Product Assembler assembles fabricated parts of small DC light sub assembly units by operating required tools. He/She also test and calibrate the assembled units to ensure the small DC based solar products meet the required quality and safety specifications. The job holder also repairs, refits and handles warranty claims for the concerned assembled units.

Personal Attributes

Solar Domestic Product Assembler is responsible for basic preparation, handling and supply of materials for Solar Domestic Products. The job holder must adhere to all safety procedures and practice good housekeeping, safe operation of specific equipment, material or processes. This is an entry level position for solar manufacturing.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4901: Introduce Renewable Energy resources](#)
2. [SGJ/N4902: Introduce Solar Energy](#)
3. [SGJ/N4903: Explain Application of Solar PV in small DC system](#)
4. [SGJ/N4904: Identify and Explain components for small solar lighting system](#)
5. [SGJ/N4905: Introduce and Explain various DC based solar products, their assembly & repair](#)
6. [SGJ/N4906: Maintain Personal Health & Safety while handling various DC based solar products](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
Country	India

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NSQF Level	2.5
Credits	9
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8212.9900
Minimum Educational Qualification & Experience	8th grade pass (and pursuing continuous schooling) with NA of experience OR 9th grade pass with NA of experience OR 5th grade pass with 4 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (1) with 1-2 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (2) with 0-6 Months of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-2.5-ES-00907-2023-V1-SCGJ
NQR Version	1.0

Remarks:

Total 270 Hours i.e. 9 Credits total duration (125 hours theory + 85 hours practical) + 30 hours of Employability Skills)+30 hours of OJT

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SGJ/N4901: Introduce Renewable Energy resources

Description

This unit is about introduce Renewable Energy resources

Scope

The scope covers the following :

- Introduction to various renewable energy Sources
- Introduction to Renewable energy Technology
- key components of the renewable energy system and outline their functions

Elements and Performance Criteria

Introduce Renewable Energy resources

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the various forms of Energy
- PC2.** discuss the various Renewable energy source and its applications and illustrate the working of various renewable energy technology with various charts and models.
- PC3.** discuss the advantage and limitations of renewable energy technologies
- PC4.** discuss the role and status of renewable energy in India
- PC5.** discuss the principle of renewable energy technologies with a focus on solar PV
- PC6.** illustrate key components of the renewable energy system and outline their functions through plant schematic
- PC7.** illustrate renewable energy technology through quiz and puzzles

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's installation policy
- KU2.** company's customer support policy
- KU3.** company's documentation policy
- KU4.** document information using appropriate corporate forms
- KU5.** obtain authorization from specified field safety officer and supervisor
- KU6.** company's reporting structure & organization culture
- KU7.** company's different department and concerned authority
- KU8.** relevant personal protective equipment's required within the manufacturing facility
- KU9.** relevant standards and regulations to be followed in the manufacturing facility
- KU10.** occupational health and safety (OHS) standards to be followed in the manufacturing facility
- KU11.** risk identification and mitigation procedure for safe working in a manufacturing facility



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare and maintain documentation
- GS2.** Read and understand training manuals in English/Hindi/ vernacular language
- GS3.** Basic understanding of vernacular/local language
- GS4.** Read user manuals and safety instructions
- GS5.** Express statements or information clearly so that others can hear and understand
- GS6.** Participate in and understand general discussions
- GS7.** Respond appropriately to any queries
- GS8.** Communicate with other employees in the organization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduce Renewable Energy resources</i>	29	21	-	-
PC1. discuss the various forms of Energy	6	-	-	-
PC2. discuss the various Renewable energy source and its applications and illustrate the working of various renewable energy technology with various charts and models.	5	5	-	-
PC3. discuss the advantage and limitations of renewable energy technologies	6	-	-	-
PC4. discuss the role and status of renewable energy in India	6	-	-	-
PC5. discuss the principle of renewable energy technologies with a focus on solar PV	6	-	-	-
PC6. illustrate key components of the renewable energy system and outline their functions through plant schematic	-	8	-	-
PC7. illustrate renewable energy technology through quiz and puzzles	-	8	-	-
NOS Total	29	21	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4901
NOS Name	Introduce Renewable Energy resources
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N4902: Introduce Solar Energy

Description

This unit is about basics of Solar Energy

Scope

The scope covers the following :

- basics of solar energy along with various fundamental concepts in electrical energy.
- the movement of sun and sun path diagram

Elements and Performance Criteria

Basics of solar energy

To be competent, the user/individual on the job must be able to:

- PC1.** explain the basics of electricity and solar energy, generation of electrical energy, thermal energy along with electrical concepts and perform simple calculations to illustrate the fundamental concepts of power and energy
- PC2.** discuss the photoelectric effect
- PC3.** explain the importance of Diffused Normal Irradiance (DNI) and Global Horizontal Irradiance (GHI) along with differences in Irradiance & Irradiation
- PC4.** discuss the basics of solar applications including solar water heater, solar cooking, solar dryer, solar concentrator, solar lamp, solar home lighting system, solar pump etc
- PC5.** discuss the movement of sun and sun path diagram and demonstrate how the movement of sun affects the solar energy generation through model illustrations, chart and videos
- PC6.** draw a schematic for generating heat and electricity from solar
- PC7.** illustrate the movement of the sun and assess its effect on the performance of the solar energy generation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The concept of the entire project
- KU2.** The entire assembly and distribution process
- KU3.** Must be well versed with the assembly timeline
- KU4.** • Time management concepts including discipline, punctuality, act in time on
• commitment and quality productive time
- KU5.** Different types of solar products
- KU6.** • Characteristics & specifications of battery, solar module, PCB (Printed Circuit Board), LED (light-Emitting-Diode) load wire and other components of a solar
• lamp as per the manufacturer
- KU7.** Structure of a basic repair and maintenance report

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare and maintain documentation
- GS2.**
 - Read and Understand training manuals in English
 - / Hindi / vernacular language
- GS3.** Basic understanding of vernacular/local language
- GS4.** Read user manuals and safety instructions
- GS5.**
 - Express statements or information clearly so that others can hear and
 - understand
- GS6.** Participate in and understand general discussions
- GS7.** Respond appropriately to any queries
- GS8.** Communicate with other employees in the organization
- GS9.** Follow organization based decision making process
- GS10.** Plan and organise work schedule to meet deadlines
- GS11.** Work constructively with others.
- GS12.** Adhere to organisation code of conduct

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of solar energy</i>	35	15	-	-
PC1. explain the basics of electricity and solar energy, generation of electrical energy, thermal energy along with electrical concepts and perform simple calculations to illustrate the fundamental concepts of power and energy	5	5	-	-
PC2. discuss the photoelectric effect	6	-	-	-
PC3. explain the importance of Diffused Normal Irradiance (DNI) and Global Horizontal Irradiance (GHI) along with differences in Irradiance & Irradiation	6	-	-	-
PC4. discuss the basics of solar applications including solar water heater, solar cooking, solar dryer, solar concentrator, solar lamp, solar home lighting system, solar pump etc	6	-	-	-
PC5. discuss the movement of sun and sun path diagram and demonstrate how the movement of sun affects the solar energy generation through model illustrations, chart and videos	6	5	-	-
PC6. draw a schematic for generating heat and electricity from solar	6	-	-	-
PC7. illustrate the movement of the sun and assess its effect on the performance of the solar energy generation.	-	5	-	-
NOS Total	35	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4902
NOS Name	Introduce Solar Energy
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N4903: Explain Application of Solar PV in small DC system

Description

This unit is about explain application of Solar PV in small DC system

Scope

The scope covers the following :

- Understand various solar DC application for lighting, power requirement and running the home appliance from solar PV panel

Elements and Performance Criteria

Explain Application of Solar PV in small DC system

To be competent, the user/individual on the job must be able to:

- PC1.** discuss in brief the application of solar energy in various DC based equipment and appliance and demonstrate the working of a solar DC system through various charts, model, video etc
- PC2.** explain the role of solar energy for small power generation and various other decentralised applications
- PC3.** discuss in detail the working of solar PV and its application in the following; mobile charging, lantern/lamps, Small Home lighting system, street light, application solar pump, etc and • Show how these applications are useful for various consumers and show their advantages (e.g. rural)
- PC4.** show how these applications are useful for various consumers and show their advantages (e.g. rural)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The concept of the entire project
- KU2.** The entire assembly and distribution process
- KU3.** Must be well versed with the assembly timeline
- KU4.** Time management concepts including discipline, punctuality, act in time on commitment and quality productive time
- KU5.** Characteristics & specifications of battery, solar module, PCB (Printed Circuit Board), LED (light-Emitting-Diode) load wire and other components of a solar lamp as per the manufacturer
- KU6.** Structure of a basic repair and maintenance report

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Prepare and maintain documentation
- GS2.** Read and Understand training manuals in english / hindi / vernacular language.
- GS3.** Basic understanding of vernacular/local language
- GS4.** Read user manuals and safety instructions
- GS5.** Express statements or information clearly so that others can hear and understand
- GS6.** Participate in and understand general discussions
- GS7.** Respond appropriately to any queries
- GS8.** Communicate with other employees in the organization
- GS9.** Follow organization based decision making process
- GS10.** Plan and organise work schedule to meet deadlines
- GS11.** Work constructively with others.
- GS12.** Adhere to organisation code of conduct
- GS13.** Recognise problems and search for solutions.
- GS14.** Choose the best methods to complete assigned tasks
- GS15.** Reduce repetition of errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Explain Application of Solar PV in small DC system</i>	30	20	-	-
PC1. discuss in brief the application of solar energy in various DC based equipment and appliance and demonstrate the working of a solar DC system through various charts, model, video etc	10	5	-	-
PC2. explain the role of solar energy for small power generation and various other decentralised applications	10	-	-	-
PC3. discuss in detail the working of solar PV and its application in the following; mobile charging, lantern/lamps, Small Home lighting system, street light, application solar pump, etc and • Show how these applications are useful for various consumers and show their advantages (e.g. rural)	10	10	-	-
PC4. show how these applications are useful for various consumers and show their advantages (e.g. rural)	-	5	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4903
NOS Name	Explain Application of Solar PV in small DC system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N4904: Identify and Explain components for small solar lighting system

Description

This unit is about identify and explain components for small solar lighting system

Scope

The scope covers the following :

- Introduction to Solar cell technology
- type of solar cell and solar module

Elements and Performance Criteria

Introduction to components for small solar lighting system

To be competent, the user/individual on the job must be able to:

- PC1.** explain the basics of Solar cell, its working and its importance and demonstrate how to identify various elements used in solar cells from the periodic table
- PC2.** explain the brief introduction to semiconductor material and its properties
- PC3.** discuss the basics of p-n junction and demonstrate the concept of p-n junction through illustrations
- PC4.** discuss the properties of Silicon and other elements for various types of solar cell fabrication and illustrate the basic fabrication of solar cell and solar module
- PC5.** discuss basics of Solar cell operation and solar module
- PC6.** discuss the basic specifications of solar PV module and demonstrate how a solar panel operates and measure basic parameters like current and voltage
- PC7.** discuss the basics of charge controller its function and applications
- PC8.** discuss the use of battery in small solar lighting system
- PC9.** discuss the basics of LED bulbs and its working and demonstrate the working of LED bulbs
- PC10.** discuss the use of basic tools like multi-meter, clamp meter, wrench, spanner, screwdriver etc and show the use of basic tools like multi-meter, clampmeter, wrench, spanner, screwdriver, etc
- PC11.** demonstrate how to identify various elements used in solar cells from the periodic table
- PC12.** demonstrate the concept of p-n junction through illustrations
- PC13.** illustrate the basic fabrication of solar cell and solar module
- PC14.** demonstrate the effect of shadow on solar module

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The concept of the entire project.

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- KU2.** The entire assembly and distribution process
- KU3.** Must be well versed with the assembly timeline
- KU4.** Time management concepts including discipline, punctuality, act in time on commitment and quality productive time
- KU5.** Different types of solar products
- KU6.** Characteristics & specifications of battery, solar module, PCB (Printed Circuit Board), LED (light-Emitting-Diode) load wire and other components of a solar lamp as per the manufacturer
- KU7.** Ability to shape and direct working/process methods according to selfdefined criteria and effective management techniques
- KU8.** Standard operating procedures for carrying out visual inspection of solar products
- KU9.** Testing procedure for verifying the performance of all the components of a solar products
- KU10.** Structure of a basic repair and maintenance report

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare and maintain documentation
- GS2.** Read and Understand training manuals in English/ Hindi / vernacular language.
- GS3.** Basic understanding of vernacular/local language
- GS4.** Read user manuals and safety instructions
- GS5.** express statements or information clearly so that others can hear and understand
- GS6.** Participate in and understand general discussions
- GS7.** Respond appropriately to any queries
- GS8.** Communicate with other employees in the organization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to components for small solar lighting system</i>	28	22	-	-
PC1. explain the basics of Solar cell, its working and its importance and demonstrate how to identify various elements used in solar cells from the periodic table	2	5	-	-
PC2. explain the brief introduction to semiconductor material and its properties	2	-	-	-
PC3. discuss the basics of p-n junction and demonstrate the concept of p-n junction through illustrations	2	3	-	-
PC4. discuss the properties of Silicon and other elements for various types of solar cell fabrication and illustrate the basic fabrication of solar cell and solar module	2	3	-	-
PC5. discuss basics of Solar cell operation and solar module	2	-	-	-
PC6. discuss the basic specifications of solar PV module and demonstrate how a solar panel operates and measure basic parameters like current and voltage	2	3	-	-
PC7. discuss the basics of charge controller its function and applications	4	-	-	-
PC8. discuss the use of battery in small solar lighting system	4	-	-	-
PC9. discuss the basics of LED bulbs and its working and demonstrate the working of LED bulbs	4	-	-	-
PC10. discuss the use of basic tools like multi-meter, clamp meter, wrench, spanner, screwdriver etc and show the use of basic tools like multi-meter, clampmeter, wrench, spanner, screwdriver, etc	4	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. demonstrate how to identify various elements used in solar cells from the periodic table	-	2	-	-
PC12. demonstrate the concept of p-n junction through illustrations	-	2	-	-
PC13. illustrate the basic fabrication of solar cell and solar module	-	2	-	-
PC14. demonstrate the effect of shadow on solar module	-	2	-	-
NOS Total	28	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4904
NOS Name	Identify and Explain components for small solar lighting system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N4905: Introduce and Explain various DC based solar products, their assembly & repair

Description

This unit is about Introduce and Explain various DC based solar products, their assembly & repair

Scope

The scope covers the following :

- various solar DC based configuration for lighting and power requirement
- Assemble and repair various Solar DC based standalone system

Elements and Performance Criteria

Introduce and Explain various DC based solar products, their assembly & repair

To be competent, the user/individual on the job must be able to:

- PC1.** discuss the configuration of various DC based standalone system and demonstrate the assembly of a solar DC system through various charts, model, video etc
- PC2.** discuss the following solar PV DC system and its assembly, Solar PV lantern/lamps, Mobile charger, home lighting system, fan and other small appliances
- PC3.** explain the wiring connection of all the standalone system and show how to connect module in series and parallel and measuring voltage and current with different combinations
- PC4.** discuss the tools and tackles required for assembly and repair of concerned system and show the use of relevant tools and tackles (e.g. soldering) and consumables
- PC5.** discuss the know-how of tools used for electrical and mechanical work of assembly of solar DC product
- PC6.** discuss how to perform testing of assembled product and show how to perform the assembly of the following; Solar PV lantern/lamps, Mobile charger, Small home lighting system
- PC7.** discuss how to perform quality check of the assembled product and demonstrate key processes like soldering of various components of solar DC system
- PC8.** discuss how to perform repair (including replacement) of the product and show how to perform repairing of the concerned system along with how to assess warranty for product replacement
- PC9.** discuss warranty and post-sale service, if any
- PC10.** discuss how to procure cost effective and quality components from various sources and show how to procure cost effective and quality components from various sources

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different types of solar domestic products
- KU2.** The concept of the entire project.

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- KU3.** The entire assembly and distribution process
- KU4.** Must be well versed with the assembly timeline
- KU5.** Time management concepts including discipline, punctuality, act in time on commitment and quality productive time
- KU6.** Characteristics & specifications of battery, solar module, PCB (Printed Circuit Board), LED (light-Emitting-Diode) load wire and other components of a solar lamp as per the manufacturer
- KU7.** Operating and working of a multimeter, soldering iron and other tools present in a tool kit
- KU8.** Preliminary concepts of solar PV system
- KU9.** Complete assembly of a solar product as per manufacturer
- KU10.** Ability to shape and direct working/process methods according to selfdefined criteria and effective management techniques
- KU11.** Testing procedure for verifying the performance of all the components of a solar product
- KU12.** Structure of a basic repair and maintenance report.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare and maintain documentation
- GS2.** Read and Understand training manuals in English / Hindi / vernacular language
- GS3.** Basic understanding of vernacular/local language
- GS4.** Read user manuals and safety instructions
- GS5.** Express statements or information clearly so that others can hear and understand
- GS6.** Participate in and understand general discussions
- GS7.** Respond appropriately to any queries
- GS8.** Communicate with other employees in the organization
- GS9.** Follow organization based decision making process
- GS10.** Plan and organise work schedule to meet deadlines
- GS11.** Work constructively with others.
- GS12.** Adhere to organisation code of conduct
- GS13.** Recognise problems and search for solutions.
- GS14.** Choose the best methods to complete assigned tasks
- GS15.** Reduce repetition of errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduce and Explain various DC based solar products, their assembly & repair</i>	28	22	-	-
PC1. discuss the configuration of various DC based standalone system and demonstrate the assembly of a solar DC system through various charts, model, video etc	2	-	-	-
PC2. discuss the following solar PV DC system and its assembly, Solar PV lantern/lamps, Mobile charger, home lighting system, fan and other small appliances	2	-	-	-
PC3. explain the wiring connection of all the standalone system and show how to connect module in series and parallel and measuring voltage and current with different combinations	2	-	-	-
PC4. discuss the tools and tackles required for assembly and repair of concerned system and show the use of relevant tools and tackles (e.g. soldering) and consumables	2	-	-	-
PC5. discuss the know-how of tools used for electrical and mechanical work of assembly of solar DC product	2	-	-	-
PC6. discuss how to perform testing of assembled product and show how to perform the assembly of the following; Solar PV lantern/lamps, Mobile charger, Small home lighting system	4	-	-	-
PC7. discuss how to perform quality check of the assembled product and demonstrate key processes like soldering of various components of solar DC system	4	-	-	-
PC8. discuss how to perform repair (including replacement) of the product and show how to perform repairing of the concerned system along with how to assess warranty for product replacement	3	-	-	-
PC9. discuss warranty and post-sale service, if any	4	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. discuss how to procure cost effective and quality components from various sources and show how to procure cost effective and quality components from various sources	3	-	-	-
NOS Total	28	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4905
NOS Name	Introduce and Explain various DC based solar products, their assembly & repair
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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SGJ/N4906: Maintain Personal Health & Safety while handling various DC based solar products

Description

This unit is about maintain Personal Health & Safety while handling various DC based solar products

Scope

The scope covers the following :

- understand the requirements for safe work area
- understand the hazards associated with solar products assembly, operations, and repair

Elements and Performance Criteria

Maintain Personal Health & Safety while handling various DC based solar products

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area and demonstrate how to keep the work area safe
- PC2.** explain the importance of administering first aid and demonstrate how to administer first aid
- PC3.** identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment and tools for ensuring safety during system assembly, repair and operations work.
- PC4.** identify the hazards associated with solar products assembly, operations, and repair and show what measures have to be taken to ensure safe practices for system assembly and repair
- PC5.** identify work safety procedures
- PC6.** explain the importance of Occupational health & Safety standards and regulations for assembly of small DC applications
- PC7.** incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The concept of the entire project
- KU2.** The entire assembly and distribution process
- KU3.** Must be well versed with the assembly timeline
- KU4.** Time management concepts including discipline, punctuality, act in time on commitment and quality productive time
- KU5.** Natural resources of energy
- KU6.** Characteristics & specifications of battery, solar module, PCB (Printed Circuit Board), LED (light-Emitting-Diode) load wire and other components of a solar lamp as per the manufacturer
- KU7.** Operating and working of a multimeter, soldering iron and other tools present in a tool kit.

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- KU8.** Preliminary concepts of solar PV system
- KU9.** Testing procedure for verifying the performance of all the components of a solar products
- KU10.** Structure of a basic repair and maintenance report.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Prepare and maintain documentation
- GS2.** Read and Understand training manuals in english/ hindi / vernacular language
- GS3.** Basic understanding of vernacular/local language
- GS4.** Read user manuals and safety instructions
- GS5.** Express statements or information clearly so that others can hear and understand
- GS6.** Participate in and understand general discussions
- GS7.** Respond appropriately to any queries
- GS8.** Communicate with other employees in the organization
- GS9.** Follow organization based decision making process
- GS10.** Plan and organise work schedule to meet deadlines
- GS11.** Work constructively with others
- GS12.** Adhere to organisation code of conduct
- GS13.** Recognise problems and search for solutions.
- GS14.** Choose the best methods to complete assigned tasks
- GS15.** Reduce repetition of errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain Personal Health & Safety while handling various DC based solar products</i>	27	23	-	-
PC1. explain the requirements for safe work area and demonstrate how to keep the work area safe	3	4	-	-
PC2. explain the importance of administering first aid and demonstrate how to administer first aid	3	4	-	-
PC3. identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment and tools for ensuring safety during system assembly, repair and operations work.	4	5	-	-
PC4. identify the hazards associated with solar products assembly, operations, and repair and show what measures have to be taken to ensure safe practices for system assembly and repair	4	5	-	-
PC5. identify work safety procedures	4	-	-	-
PC6. explain the importance of Occupational health & Safety standards and regulations for assembly of small DC applications	5	-	-	-
PC7. incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices.	4	5	-	-
NOS Total	27	23	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4906
NOS Name	Maintain Personal Health & Safety while handling various DC based solar products
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Assembler
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4901.Introduce Renewable Energy resources	29	21	0	0	50	14
SGJ/N4902.Introduce Solar Energy	35	15	0	0	50	14
SGJ/N4903.Explain Application of Solar PV in small DC system	30	20	0	0	50	14
SGJ/N4904.Identify and Explain components for small solar lighting system	28	22	0	0	50	14
SGJ/N4905.Introduce and Explain various DC based solar products, their assembly & repair	28	22	0	0	50	14
SGJ/N4906.Maintain Personal Health & Safety while handling various DC based solar products	27	23	0	0	50	14
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	16
Total	197	153	-	-	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.